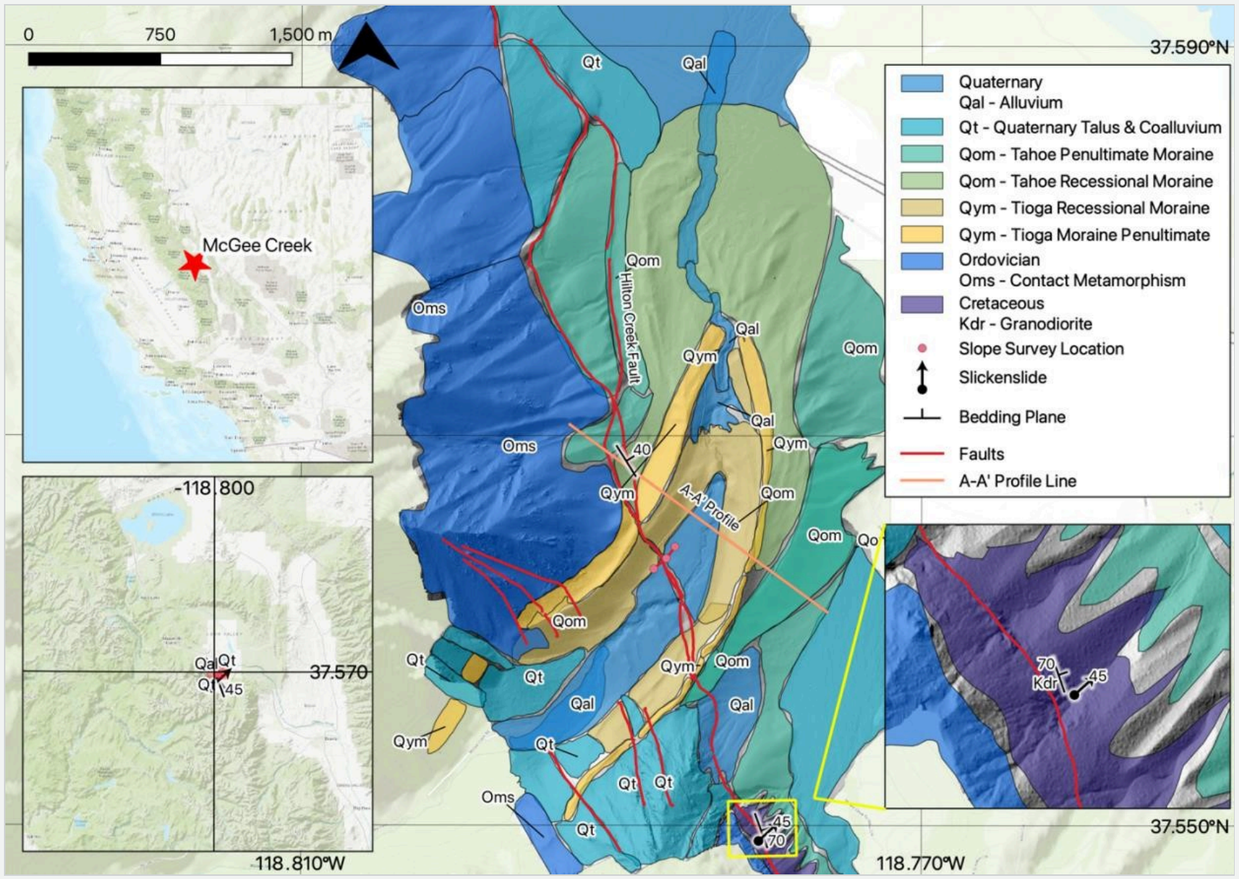
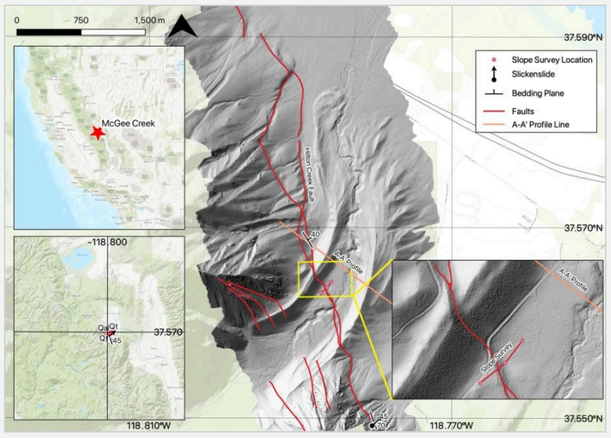
Faulted moraines of McGee Creek

Slip on the Hilton Creek Fault and the relative ages of the Tioga and Tahoe glaciations, eastern Sierra Nevada

Nick Anderson
EPS 118 field report
June 2022

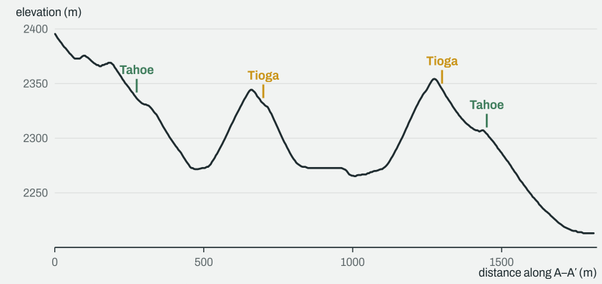


Surficial deposits mapped in the field. Tioga moraines (yellows, 18.8 ± 1.9 ka) sit inside the older Tahoe moraines (greens, 144 ± 14 ka) Also mapped: Cretaceous granodiorite, Ordovician metasedimentary rock, talus and alluvium. Red lines are faults; inset shows the slope-survey site.



Relief map: faults, the A–A′ profile line and the measured slope profile across the Hilton Creek scarp.

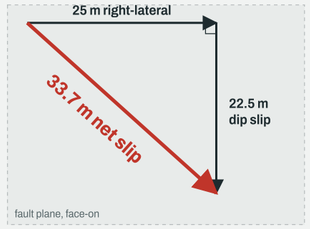
A–A′ topographic profile across both moraines

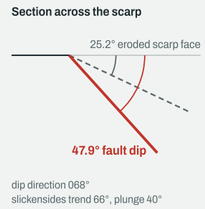


Small rounded Tahoe hills at ~275 and ~1450 m; large angular Tioga crests at ~700 and ~1300 m. Slopes reach ±40° on the Tioga crests and ±20° on the Tahoe hills.

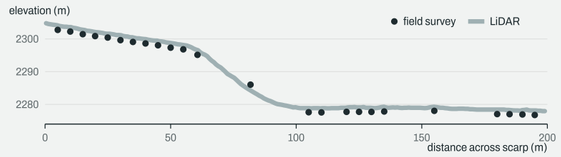
33.7 m of oblique slip since the last glacial maximum

Slip rate through the Tioga moraine: **≈ 1.77 mm/yr** over ~19 kyr





Scarp profile: field survey against LiDAR



The hand survey and the LiDAR DEM agree along 200 m across the scarp. Across the Tioga moraine the scarp records 9.6 m of vertical offset; its eroded face slopes 25.2° today. Fitting the fault plane to the DEM (Qprof) gives a dip of 47.9° toward 068°; combined with the elevation change along the scarp, that gives 22.5 m of dip slip. LiDAR shows ~25 m of lateral offset.

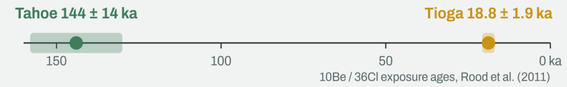
Setting

McGee Creek is a U-shaped glacial valley on the eastern Sierra front, south of Long Valley Caldera. Cretaceous Round Valley Peak granodiorite intruded Ordovician sediments, baking them to hornfels, marble and slate; Tahoe (MIS 4–6) and Tioga (MIS 2) glaciers then carved the valley and left moraines of till. The 22 km Hilton Creek Fault, an oblique normal range-front fault of the Walker Lane system, cuts those moraines. Earlier work proposed 7–9 events of 3–4 m in the last 20–25 kyr and ~1.1 mm/yr over the last 16 kyr.

Methods

Faults, slickensides and deposit contacts mapped in FieldMove Clino. A 200 m slope survey across the scarp with sighting level, Brunton, tape and stadia rod, checked against a LiDAR DEM. Clast counts on both moraines for lithology, size, oxidation, burial and xenolith protrusion using a hand lens, meter stick and dilute HCl. Fault-plane fit, offsets and slip computed in QGIS (Qprof) and Python.

Tioga moraines are younger than the Tahoe moraines, even without exposure dates

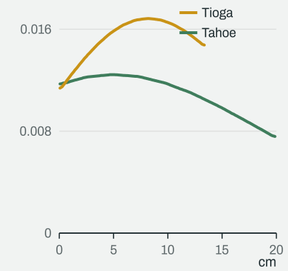


- **Crosscutting.** Sharp Tioga crests cut across the eroded Tahoe moraines on the map.
- **Crest shape.** Tioga crests slope up to ±40°; the rounded Tahoe hills only ±20°.
- **Clast size.** Tioga clasts reach 250 cm² and more. Tahoe clasts average ~6.25 cm² and top out near 175 cm². Tahoe igneous clasts reach ~225 cm long but mostly stay under 30 cm; most Tahoe metasedimentary clasts are under 2.5 cm and heavily oxidized.
- **Weathering and burial.** Diorite xenoliths stand out of Tioga granodiorite boulders by less than 5 cm, but more often by over 5 cm on Tahoe boulders. Tioga boulders expose more of their short (z) axis, ~8 cm against ~5 cm, so Tahoe boulders are more deeply buried.

Xenolith protrusion



Exposed boulder z-axis



Conclusion

Crosscutting, crest shape and weathering all put the Tioga moraines after the Tahoe moraines, matching the exposure ages. Slip through the glacial deposits is oblique normal with a right-lateral component (slickensides trend 66°, plunge 40°), at ~1.77 mm/yr. Weak, unconsolidated till along the fault plane could potentially host an earthquake of M6.8.

Berry (1997) Geomorphology 20, 177–195. Blackwelder (1931) GSA Bulletin 42, 865–922. Hildreth, Fierstein & Calvert (2018) Quat. Sci. Rev. 198, 242–265. Rood, Burbank & Finkel (2011) Quat. Sci. Rev. 30, 646–661. Unruh, Humphrey & Barron (2003) Geology 31, 327–330. Wahrhaftig et al. (2019) USGS SIM 3414.